VWAP Playbook for CME Equity Index Futures

Advanced-to-expert guideline for ES / NQ / MES / MNQ traders

The big idea
In index futures, VWAP works best as a stack of trader-defined session lines rather than one all-day average. Because ES and NQ trade nearly around the clock on CME Globex, the real edge comes from reading handoffs between Globex inventory, Europe, U.S. macro releases, the cash open, and the close.

What changes from the generic VWAP playbook?
Futures extended hours are not the same as equity extended hours. Liquidity can still thin overnight, but the order book is centralized and the overnight tape is part of the primary auction rather than a side market. That makes overnight VWAP much more usable in ES/NQ than premarket VWAP is in many individual stocks.

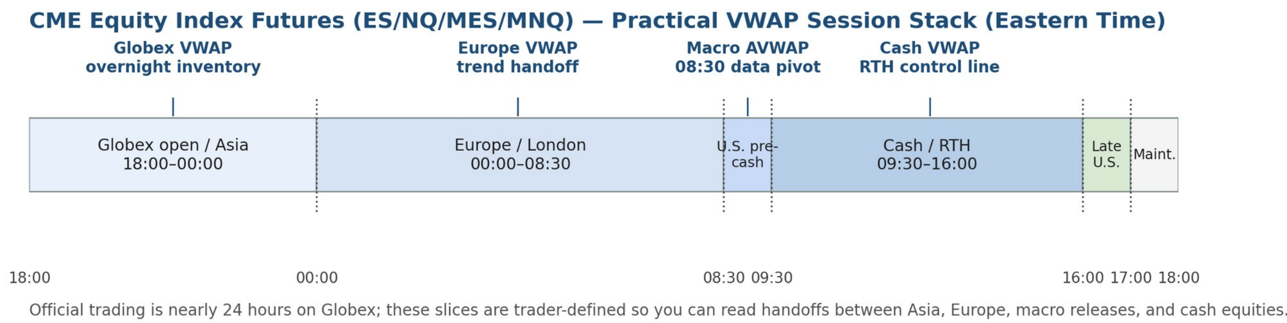

Figure 1. Session-specific VWAP map used by this playbook. These slices are practical trading windows, not a promise that your platform will calculate them automatically.

How to use this version

- Treat the overnight Globex VWAP as an inventory and acceptance map, not as an automatic fade line.
- Use event AVWAP aggressively on 08:30 ET data days and 14:00/14:30 ET Fed days; those are the cleanest volatility pivots in futures.
- Separate trader-defined slices from exchange-defined hours: the exchange is continuous, but your decision model should not be.
- Demand acceptance above or below the line stack. The cross gives context; the retest is the trade.

1. Session stack and line hierarchy

For CME equity index futures, the exchange-defined trading day is nearly 24 hours on Globex. The official schedule for ES and NQ is Sunday 6:00 p.m. ET to Friday 5:00 p.m. ET with a daily maintenance period from 5:00 p.m. to 6:00 p.m. ET. In practice, advanced traders carve that stream into decision windows that line up with real changes in participation and information flow: Asia, Europe, U.S. macro, the cash session, and the late U.S. close.

Line	Reset / anchor	What it measures	Reliability
Globex VWAP	18:00 ET reopen	Overnight inventory basis and acceptance away from prior settle	High
Europe VWAP	00:00 ET	Whether Europe is extending or fading the Asia move	Med–High
Macro AVWAP	08:30 ET data bar	Fresh cost basis after CPI / NFP / PPI / claims-type shocks	Very high
Cash VWAP	09:30 ET cash open	RTH control line once U.S. cash participation takes over	Very high
Fed AVWAP	14:00 / 14:30 ET	Policy shock cost basis on FOMC days	Very high
Weekly VWAP	Week open	Higher-timeframe location and trend pressure	High

Expert rule
Your base-case long is not “price above VWAP.” It is: macro or session handoff created a higher cost basis, price accepted above the fast line, and the retest held while the slower line was aligned or flat-to-rising.

- Best long environment: Globex VWAP up, Europe VWAP up or flat, price above weekly VWAP, then a clean cash-open retest.
- Best short environment: overnight squeeze fails into cash, price loses the overnight stack, and RTH VWAP cannot reclaim.
- Do not overcomplicate the stack. Most days you only need three active references: overnight VWAP, current RTH VWAP, and one event AVWAP.

2. Core playbook setups

A. Overnight inventory handoff

Bullish version: overnight price holds above Globex VWAP, Europe does not fully unwind it, and the first RTH pullback holds above both Globex VWAP and a rising cash VWAP. Entry comes on the retest hold. Invalidation is a cash-session failure back through both lines.

B. Cash-open rejection

Bearish version: price arrives above overnight VWAP into 09:30 but cannot hold the opening drive, loses the overnight stack, and fails the first reclaim from underneath. That often signals overnight inventory was offside and is being unwound.

C. Weekly + session alignment

This is the highest-quality continuation pattern. Weekly VWAP, overnight VWAP, and current-session VWAP stack in the same direction. In that regime, fading +2σ extensions too early is expensive; the better trade is buying or selling pullbacks into the stack.

D. Rotation day logic

If cash VWAP is flat by late morning and both sides of it have traded multiple times, downgrade trend expectations. Use ±1σ and ±2σ as mean-reversion targets back toward cash VWAP rather than forcing breakout continuation.

Setups are ranked by quality when three conditions line up: slope, acceptance, and timing.

Setup	Trigger	Entry logic	Primary target
Overnight handoff	Pullback holds stacked VWAPs	Reclaim bar or first higher low	ONH / prior swing / +1σ
Cash-open rejection	Lose overnight VWAP then fail reclaim	Short the failed retest	Overnight midpoint / ONL
Macro pivot	Event AVWAP holds on first controlled retest	Enter away from AVWAP	Impulsive leg extension
Rotation day fade	Outer band rejects while VWAP stays flat	Fade back toward VWAP	Day VWAP

3. Volatility pivots and anchored VWAP

Volatility pivots are where futures traders gain the biggest advantage from AVWAP. Use a fresh anchor when the market receives information that plausibly resets value: 08:30 ET economic data, 10:00 ET ISM / consumer data, 14:00 ET FOMC statements, 14:30 ET Powell press conferences, or a shock move at the cash open.

- Do not anchor to random candles. Anchor to the first bar that clearly changes the auction.
- After the first impulse, wait for the first controlled retest; the spike itself is not the setup.
- A clean continuation has price holding above event AVWAP and then building a new impulse while cash VWAP catches up.
- A pivot-failure reversal happens when price loses event AVWAP, crosses cash VWAP, retests the cluster from the wrong side, and rejects.

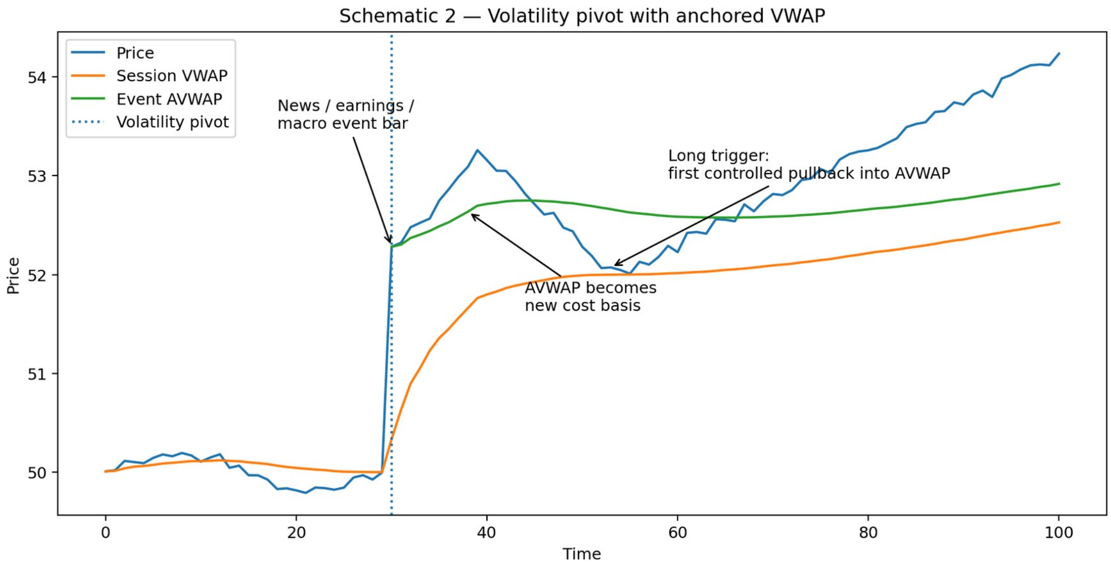

Figure 2. Event AVWAP becomes the new cost basis after a real volatility shock. The long is the first controlled retest, not the spike.

High-volatility protocol
On CPI and FOMC days, I favor continuation over fade until a second balance forms. That means smaller mean-reversion size, wider structural stops, and more patience for the first post-event pullback.

4. Crossing strategies: price/VWAP and line/VWAP

Crosses are information, not execution by themselves. The execution decision comes from acceptance after the cross. That rule matters even more when using fast and slow VWAPs because these lines lag and can whip in midday chop.

Cross type	Bullish read	Short read	What confirms it
Price vs cash VWAP	Reclaim, hold, then build higher lows above VWAP	Lose, fail reclaim, then build lower highs below VWAP	VWAP slope and a clean retest
Cash VWAP vs Globex VWAP	RTH builds a higher cost basis than the overnight session	RTH cannot hold above overnight inventory and rolls back below it	Price accepted above or below both
Event AVWAP vs weekly VWAP	Fresh information is strong enough to move the market above HTF value	Event reprice cannot sustain above HTF value	Retest of the crossover zone
Multiple touches	Usually noise unless slope and timing improve	Usually noise unless slope and timing improve	Skip or wait for a reset

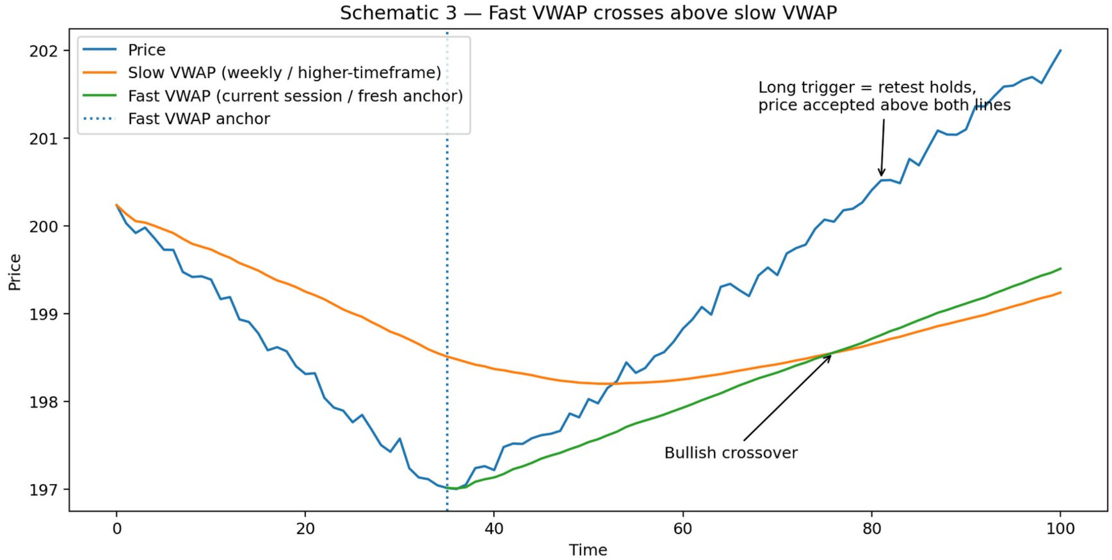

Figure 3. A fast VWAP crossing above a slower VWAP becomes actionable only when price is accepted above both lines on the retest.

- Best long cross: fast line up through slow line, price above both, fast slope positive, and the retest occurs before lunch chop or after a clean volatility pivot.
- Best short cross: fast line down through slow line, price below both, retest fails, and the slower line is flat-to-down rather than still rising hard.
- Avoid raw crosses inside a flat value area. If the tape has already crossed the same lines three times, you do not have a new signal.

5. Trend day vs rotation day

Your VWAP framework changes with the day type. Rising VWAPs, one-sided tape, and shallow pullbacks favor trend logic. Flat VWAP, repeated crossings, and rejected outer bands favor rotation logic.

Day type	VWAP behavior	Trade plan	Avoid
Trend day	VWAP rises or falls with shallow pullbacks	Buy/sell pullbacks into VWAP or AVWAP; trail runner on fast line	Fading first extension without rejection
Rotation day	VWAP flat; both sides trade repeatedly	Fade ±1σ / ±2σ back toward VWAP	Breakout trades in lunch chop
Event trend	Event AVWAP and cash VWAP reprice sharply	Use event AVWAP as the primary hold/fail line	Oversized contrarian fades

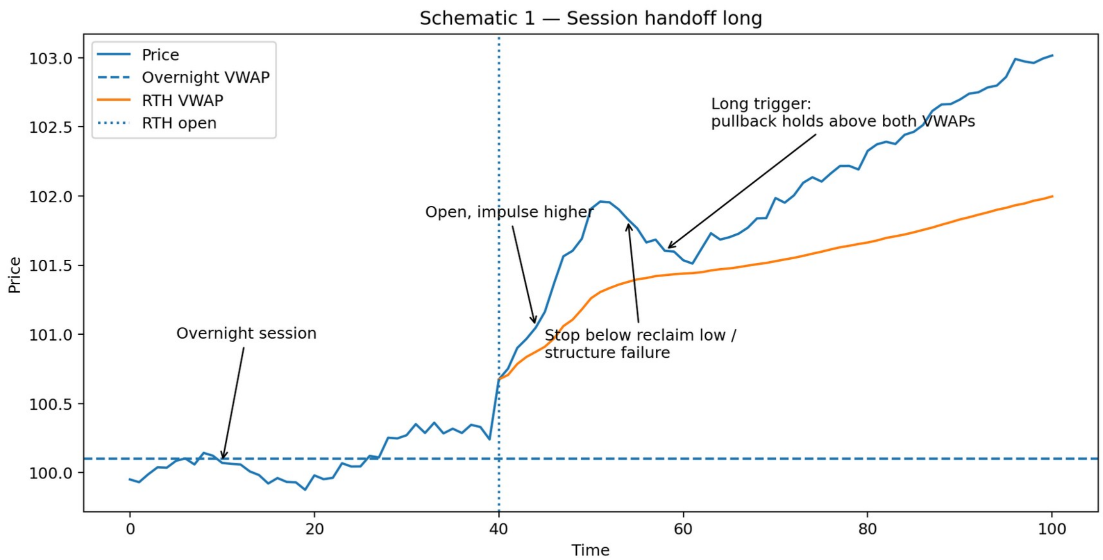

Figure 4. Session-handoff continuation: the higher-quality long comes from a pullback that holds the stacked lines, not from the opening spike.

6. Daily routine and trade construction

Time block	What to map	Main question
18:00 reopen	Gap vs prior settle, Globex VWAP	Is overnight inventory building in one direction?
00:00 Europe	Europe VWAP, prior balance edges	Is Europe extending or fading Asia?
08:30 macro	Event AVWAP, initial impulse	Did data create a new cost basis or just noise?
09:30 cash	Cash VWAP, ONH/ONL, opening drive	Continuation or inventory correction?
11:00–13:30	VWAP slope, band behavior	Trend continuation or rotation regime?
14:00–14:30	Fed AVWAP on policy days	Fresh regime shift or failed repricing?
15:30–16:00	Close behavior vs cash VWAP	Directional close or late flow distortion?

Trade construction template

Long: identify the active reference line, wait for acceptance above it, enter on the retest, place the stop below the reclaim low or AVWAP failure, take partials at structure or +1σ, and trail the runner under the fast line. Invert the logic for shorts.

Example: CPI continuation long

08:30 ET CPI creates an impulse higher. Price pulls back into event AVWAP and stays above overnight VWAP. Enter on the first reversal candle away from AVWAP. Target prior spike high, then trail a runner under event AVWAP.

Example: overnight squeeze failure short

Overnight rally cannot hold above Globex VWAP after the cash open. Price loses the stack, retests from underneath, and fails. Short the rejection for a move back toward overnight midpoint or low.

7. Risk controls, common mistakes, and source notes

- Reduce size or flatten into scheduled 08:30 and 14:00/14:30 event risk unless the event itself is your setup.
- Use structural stops, not arbitrary tick counts. VWAP failure without structure context is too noisy.
- Do not treat overnight VWAP the same on every day. On macro or geopolitical shock days it matters more; on dead overnight sessions it matters less.
- Do not force line crosses in lunch chop. A flat VWAP with repeated recrossing is a skip, not an opportunity.
- Journal by setup family: overnight handoff, macro AVWAP, cash-open rejection, weekly alignment, rotation fade.

Most expensive futures mistake
Using a beautiful overnight trend to justify a bad cash-open entry. The handoff is the edge. If the cash session rejects the overnight value stack, the overnight trend is not your friend anymore.

Selected sources

- **CME Group — E-mini Nasdaq-100 Futures Contract Specs —** source — Official Globex hours and maintenance window.
- **CME Group — E-mini S&P 500 Futures Overview —** source — Notes nearly 24-hour access and ES’s role versus cash equities.
- **Nasdaq — Market Activity / Market Hours —** source — Used here for the U.S. cash session open and close.
- **Gao, Han, Li, Zhou (2018), Market Intraday Momentum —** source — First half-hour return predicts the last half-hour return, stronger on volatile and high-volume days.
- **Biais, Glosten, Spatt (2005), Market Microstructure: A Survey —** source — General microstructure background for volume, liquidity, and price discovery.